

Que 1

```
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam create-group
--group-name SoftwareEngineering

{
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPATC2FAR6CKORTSBUW6",
    "Arn": "arn:aws:iam::212208750468:group/SoftwareEngineering",
    "CreateDate": "2026-01-19T07:43:22+00:00"
  }
}
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ █

}
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam create-user --
-user-name irraj
{
  "User": {
    "Path": "/",
    "UserName": "irraj",
    "UserId": "AIDATC2FAR6CMHNIFWUDA",
    "Arn": "arn:aws:iam::212208750468:user/irraj",
    "CreateDate": "2026-01-19T07:45:53+00:00"
  }
}
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ █

}
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam get-user --u
er-name irraj
{
  "User": {
    "Path": "/",
    "UserName": "irraj",
    "UserId": "AIDATC2FAR6CMHNIFWUDA",
    "Arn": "arn:aws:iam::212208750468:user/irraj",
    "CreateDate": "2026-01-19T07:45:53+00:00"
  }
}
}
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam add-user-to
roup \
--user-name irraj \
--group-name SoftwareEngineering
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ █
```

```

● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam get-group --group-name SoftwareEngineering
{
  "Users": [
    {
      "Path": "/",
      "UserName": "irraj",
      "UserId": "AIDATC2FAR6CMHNIFWUDA",
      "Arn": "arn:aws:iam::212208750468:user/irraj",
      "CreateDate": "2026-01-19T07:45:53+00:00"
    }
  ],
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPATC2FAR6CKORTSBUW6",
    "Arn": "arn:aws:iam::212208750468:group/SoftwareEngineering",
    "CreateDate": "2026-01-19T07:43:22+00:00"
  }
}
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ █

```

```

}
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam list-policies --scope AWS | grep AdministratorAccess
  "PolicyName": "AdministratorAccess",
  "Arn": "arn:aws:iam::aws:policy/AdministratorAccess",
  "PolicyName": "AdministratorAccess-Amplify",
  "Arn": "arn:aws:iam::aws:policy/AdministratorAccess-Amplify",
  "PolicyName": "AWSAuditManagerAdministratorAccess",
  "Arn": "arn:aws:iam::aws:policy/AWSAuditManagerAdministratorAccess",
  "PolicyName": "AdministratorAccess-AWSElasticBeanstalk",
  "Arn": "arn:aws:iam::aws:policy/AdministratorAccess-AWSElasticBeanstalk",
  "PolicyName": "AWSManagementConsoleAdministratorAccess",
  "Arn": "arn:aws:iam::aws:policy/job-function/AWSManagementConsoleAdministratorAccess",
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ █

```

```

○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ ^C
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam attach-group-policy \
  --group-name SoftwareEngineering \
  --policy-arn arn:aws:iam::aws:policy/AdministratorAccess
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ aws iam list-attached-group-policies \
> █

```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
~
~
~
~
```

```
variable "subnet_cidr_block" {
  description = "CIDR block for the subnet"
  type        = string
}

variable "availability_zone" {
  description = "AWS availability zone"
  type        = string
}

variable "env_prefix" {
  description = "Prefix for tagging resources"
  type        = string
}

variable "instance_type" {
  description = "EC2 instance type"
  type        = string
}

-- INSERT --
```

```

variable "subnet_cidr_block" {
  description = "CIDR block for the subnet"
  type        = string
}

variable "availability_zone" {
  description = "AWS availability zone"
  type        = string
}

variable "env_prefix" {
  description = "Prefix for tagging resources"
  type        = string
}

variable "instance_type" {
  description = "EC2 instance type"
  type        = string
}

```

"variables.tf" 30L, 607B

29,1

terraform -v

```

5
6 resource "aws_vpc" "myapp_vpc" {
7   cidr_block = var.vpc_cidr_block
8
9   tags = {
10     Name = "${var.env_prefix}-vpc"
11   }
12 }
13
14 resource "aws_subnet" "myapp_subnet" {
15   vpc_id            = aws_vpc.myapp_vpc.id
16   cidr_block         = var.subnet_cidr_block
17   availability_zone = var.availability_zone
18
19   tags = {
20     Name = "${var.env_prefix}-subnet-1"
21   }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

resource "aws_default_route_table" "rt" {
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

```

```

}
}

```

```

resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.myapp_vpc.id

  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

data "http" "myip" {
  url = "https://icanhazip.com"
}

locals {
  my_ip = "${chomp(data.http.myip.response_body)}/32"
}

```

```
32 resource "aws_default_route_table" "rt" {  
42     }  
43 }  
44 resource "aws_default_security_group" "default_sg" {  
45     vpc_id = aws_vpc.myapp_vpc.id  
46  
47     ingress {  
48         from_port    = 22  
49         to_port      = 22  
50         protocol     = "tcp"  
51         cidr_blocks = [local.my_ip]  
52     }  
53  
54     ingress {  
55         from_port    = 80  
56         to_port      = 80  
57         protocol     = "tcp"
```

```
ingress {
  from_port    = 80
  to_port      = 80
  protocol     = "tcp"
  cidr_blocks  = ["0.0.0.0/0"]
}
```

```
ingress {
  from_port    = 443
  to_port      = 443
  protocol     = "tcp"
  cidr_blocks  = ["0.0.0.0/0"]
}
```

```
resource aws_default_security_group default_sg {
```

```
  egress {
    from_port    = 0
    to_port      = 0
    protocol     = "-1"
    cidr_blocks  = ["0.0.0.0/0"]
  }
```

```
  tags = {
    Name = "${var.env_prefix}-default-sg"
  }
```

```
}
```

```
}
```

```
resource "aws_key_pair" "serverkey" {
  key_name     = "serverkey"
  public_key   = file("~/ssh/id_ed25519.pub")
}
```

```
  Name = "${var.env_prefix}-ec2-instance"
}
```

```
}
```

```
output "ec2_public_ip" {
  value = aws_instance.myapp.public_ip
}
```



```

}

resource "aws_instance" "myapp" {
  ami                = "ami-0d4f9e3b7a25a12f5" # Amazon Linux 2023
  instance_type      = var.instance_type
  subnet_id          = aws_subnet.myapp_subnet.id
  vpc_security_group_ids = [aws_default_security_group.default_sg.id]
  availability_zone   = var.availability_zone
  associate_public_ip_address = true
  key_name            = aws_key_pair.serverkey.key_name
  user_data           = file("entry-script.sh")

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

```

```

GNU nano 7.2                                entry-script.sh *
#!/bin/bash
yum update -y
yum install -y nginx openssl

mkdir -p /etc/nginx/ssl

openssl req -x509 -nodes -days 365 \
  -newkey rsa:2048 \
  -keyout /etc/nginx/ssl/nginx.key \
  -out /etc/nginx/ssl/nginx.crt \
  -subj "/CN=localhost"

cat <<EOF > /usr/share/nginx/html/index.html
<h1>This is Samreen's Terraform environment</h1>
EOF

```

```

vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix           = "dev"
instance_type        = "t3.micro"

```

@23-22411-027-max → /workspaces/cloud-lab/Labexam (main) \$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""

```

|      .  *o .  E|
|      S o . = .  |
|      .. = + o . = |
|      o ** + B o . |
|      . = . B . . = |
|      . * o . . + |
+-----[SHA256]-----+

```

```
+-----[SHA256]-----+
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ ls ~/.ssh
id_ed25519 id_ed25519.pub
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $
```

```
@23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ terraform init
```

```
● @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/http...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future
```

```
@23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ terraform init
terraform plan
terraform apply -auto-approve
terraform init
terraform plan ...
```

```
    + "Name" = "dev-vpc"
  }
+ tags_all                                     = {
  + "Name" = "dev-vpc"
  }
}
```

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ ec2_public_ip = (known after apply)
```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if run "terraform apply" now.

```
@23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ terraform init
terraform plan
terraform apply -auto-approve
terraform init ...
```

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
ec2_public_ip = "158.252.77.8"
```

```
○ @23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $
```

← → ↻ ⚠ Not secure 158.252.77.8

This is Samreen's Terraform environment

```
aws_instance.myapp: Still modifying... [id=i-01afa7398d951ec46, 01m10s elapsed]
aws_instance.myapp: Modifications complete after 1m15s [id=i-01afa7398d951ec46]
```

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Outputs:

```
ec2_public_ip = "51.112.167.238"
```

```
@23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ nano entry-script.sh
```

```
@23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $
```

← → ↻ ⚠ Not secure 51.112.167.238

This is Samreen's Terraform environment

```
@23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $ terraform destroy
```

```
aws_key_pair.serverkey: Destroying... [id=serverkey]
```

```
aws_default_security_group.default_sg: Destruction complete after 0s
```

```
aws_subnet.myapp_subnet: Destroying... [id=subnet-0240a59f938cc605e]
```

```
aws_key_pair.serverkey: Destruction complete after 0s
```

```
aws_subnet.myapp_subnet: Destruction complete after 1s
```

```
aws_vpc.myapp_vpc: Destroying... [id=vpc-0d474e52200fd59f9]
```

```
aws_vpc.myapp_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 7 destroyed.

```
@23-22411-027-max →/workspaces/cloud-lab/Labexam (main) $
```

Ques 3

GNU nano 7.2

hosts

[ec2]

51.112.253.167

[ec2:vars]

ansible_user=ec2-user

ansible_ssh_private_key_file=~/.ssh/id_ed25519

ansible_ssh_common_args='-o StrictHostKeyChecking=no'

15

GNU nano 7.2

ansible.cfg *

[defaults]

inventory = ./hosts

host_key_checking = False

interpreter_python = /usr/bin/python3

GNU nano 7.2 my-playbook.yml *

enabled: no

ignore_errors: yes

- name: Remove nginx package

yum:

name: nginx

state: absent

ignore_errors: yes

- name: Install httpd

yum:

name: httpd

state: present

- name: Start and enable httpd

service:

bas
nar

```
@23-22411-027-max /.../cloud-lab/Labexam/Lab_exam/ansible (main) $ nano my-playbook.yml
@23-22411-027-max →.../cloud-lab/Labexam/Lab_exam/ansible (main) $ ansible-playbook my-playbook.yml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/cloud-lab/Labexam/Lab_exam/ansible), :
it as an ansible.cfg source. For more information see
https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not i
'all'
[WARNING]: Could not match supplied host pattern, ignoring: ec2

PLAY [Configure Apache Web Server on EC2] *****
skipping: no hosts matched

PLAY RECAP *****
```



This is Samreen's Terraform environment

<https://github.com/23-22411-027-max/cloud-lab.git>